

#Task 3

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Analyze Network Traffic by Using Wireshark, Nmap

Introduction

In this task, I used Wireshark to gain hands-on experience in monitoring and analyzing the data (traffic) that moves through the network while browsing the internet. The goal of this task is to help me learn how to observe the connection between my device and the websites I visit, and to understand the protocols used, such as TCP, UDP, DNS, and HTTP.

Through this experience, I started to understand how data moves, what it looks like, and how to tell the difference between normal packets and ones that may show a problem like delay or a dropped connection. I also learned how to use the Nmap tool to scan open ports on websites, which is helpful in checking network or website security.

This task helped me understand the basics of network analysis, and it's an important first step in learning cybersecurity.

- 1) At the beginning, I downloaded the Wireshark program.



- 2) I have completed the Live Data Capture step, and while capturing, I generated data traffic by performing three of the required activities.

- 3) I found the following protocols during the packet capture, and here are some simple definitions about them.

- **TCP (Transmission Control Protocol):**

A connection-oriented protocol that ensures reliable and ordered delivery of data between devices. It checks for errors and guarantees that all data arrives intact.

- **UDP (User Datagram Protocol):**

A connectionless protocol that allows fast data transfer without guaranteeing delivery. It is commonly used in applications requiring speed, such as live streaming, DNS, and the QUIC protocol.

- **DNS (Domain Name System):**

This system translates human-readable domain names (like google.com) into IP addresses that computers use to communicate over the internet.

- **ARP (Address Resolution Protocol):**

Used to map IP addresses to physical MAC addresses within a local network, enabling devices to identify each other at the hardware level.

- **HTTP (Hypertext Transfer Protocol):**

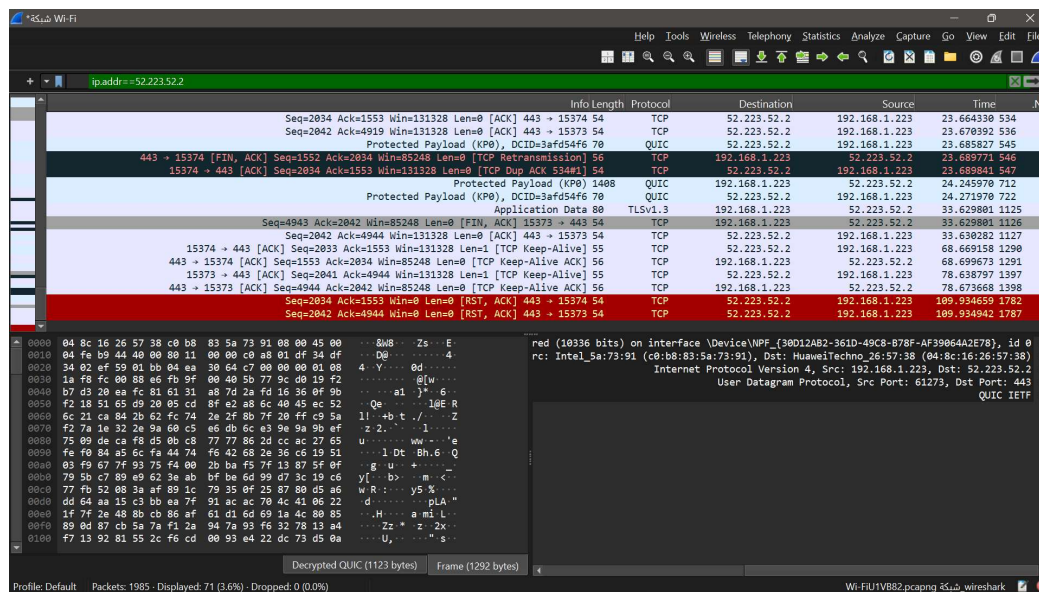
The foundation of data communication on the web, used by browsers to request and receive web pages from servers.

- QUIC (Quick UDP Internet Connections):

A modern transport protocol built on UDP, designed to provide faster, more secure, and reliable connections for web browsing. Developed by Google.

4) Analyze One Selected Packet in screenshot and Detect and Analyze Packet Issues:

1-www.afaawqetaan.com



--Analyze One Selected Packet:

- Packet Number: 534
- Protocol Type: TCP
- Source IP Address: 192.168.1.223
- Destination IP Address: 52.223.52.2
- Source Port: 15374
- Destination Port: 443

Description:

This packet is a TCP acknowledgment (ACK). It means the user's device told the server that it got the data and is ready to receive more.

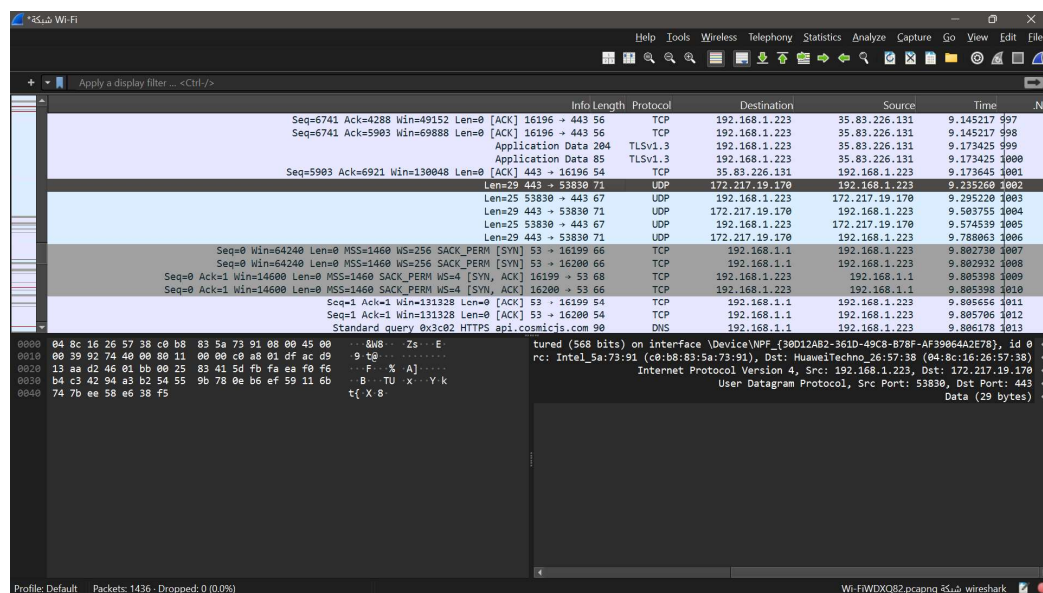
Technical Notes:

There was no delay or resend in this packet. This is normal in TCP communication to make sure data is sent correctly between the device and the server using HTTPS.

–Detect and Analyze Packet Issues :

- **Number of affected packets:** 1 selected (TCP Reset) other effected 22 red packets .
 - **Possible cause:** Connection reset by the client, possibly due to timeout, app error, or network issue.
 - **Impact on connection:** Connection with the server was abruptly terminated.
 - **Suggested solutions:** Check internet stability, restart router, or try using a wired connection.
- **Wireshark Colors :**
 - Red – indicates a connection issue
 - Red packets : 22
 - Yellow packets : 44

2- www.wetaan.com



–Analyze One Selected Packet:

- **Packet Number:** 1002
- **Protocol Type:** UDP
- **Source IP Address:** 192.168.1.223
- **Destination IP Address:** 172.217.19.170
- **Source Port:** 53830
- **Destination Port:** 443

Description:

This is a small UDP packet sent from the device to a server. It is probably part of a QUIC connection which uses UDP for HTTPS on port 443.

Technical Notes:

There was no retransmission or delay in this packet. UDP does not guarantee delivery, so a higher protocol like QUIC handles making sure the data arrives correctly.

--Detect and Analyze Packet Issues :

Seq=1 Ack=281 Win=269568 Len=0 [ACK] 16224 → 80 56	TCP	192.168.1.223	142.250.181.67	56.167114	1394
HTTP/1.1 304 Not Modified 277	HTTP	192.168.1.223	142.250.181.67	56.167114	1395

- Packet Number: 1395
- Protocol Type: HTTP
- Number of affected packets: 1 selected (yellow), no retransmission observed.
- Possible cause: Cached response triggered yellow marking.
- Impact on connection: No negative effect; response was valid.
- Suggested solutions: No action needed – expected caching behavior.
- Wireshark Colors:
 - Yellow/Orange = Warning (delay)

Red packets : 10 effected

Yellow packets :35 effected

3- <https://mashaalalkhair.org.sa/>

Seq=26 Ack=25 Win=0 Len=0 [RST, ACK] 443 → 16711 54	TCP	86.51.81.96	192.168.1.223	111.471693	1476
Standard query response 0xd0fd A stream-production.avcdn.net CNAME stream-production.avcdn 354	DNS	192.168.1.223	192.168.1.1	111.471578	1475

--Analyze One Selected Packet:

- Packet Number: 1475
- Protocol Type: DNS
- Source IP Address: 192.168.1.1
- Destination IP Address: 192.168.1.223
- Source Port: 53
- Destination Port: (Temporary port on device)

Description:

This packet is a DNS response for the website *stream-production.avcdn.net*. It shows the website's address by giving several IP addresses through something called a CNAME chain.

Technical Notes:

There are no problems with this packet. It's colored light blue, which means it is normal DNS traffic.

--Detect and Analyze Packet Issues:

Seq=1 ACK=25 Win=301 Len=0 [ACK] 10711 → 443 96	TCP	192.168.1.223	86.51.81.96	111.471170 1471
Application Data 78	TLSv1.2	192.168.1.223	86.51.81.96	111.471170 1472
Seq=25 Ack=26 Win=501 Len=0 [RST, ACK] 16711 → 443 56	TCP	192.168.1.223	86.51.81.96	111.471170 1473
16711 → 443 [ACK] Seq=26 Ack=1 Win=513 Len=0 [TCP Dup ACK 1468#1] 84	TCP	86.51.81.96	192.168.1.223	111.471475 1474
...Standard query response 0xd6fd A stream-production.avcdn.net CNAME stream-production.avcdn 354	DNS	192.168.1.223	192.168.1.1	111.471670 1475

- **Packet Number:** 1473
 - **Number of affected packets:** 1 (TCP Reset) red effected
 - **Possible cause:** Server closed the connection suddenly (timeout or rejection)
 - **Impact on connection:** Connection was interrupted
 - **Suggested solutions:** Check internet, reload page, restart router if needed
 - **Wireshark Colors :**
 - Red – indicates a connection issue
- Red packets : 20 -Yellow packets: 28

• Sources:

- Cisco:

What is TCP?

What is ARP?

- Cloudflare:

What is UDP?

What is DNS?

What is QUIC?

- MDN Web Docs:

[What is HTTP?](#)

Part 2 :

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.22000.2538]
(c) Microsoft Corporation. All rights reserved.

C:\Program Files (x86)\Nmap>nmap --version
Nmap version 7.97 ( https://nmap.org )
Platform: i686-pc-windows-windows
Compiled with: nmap-liblua-5.4.7 openssl-3.0.16 nmap-libssh2-1.11.1 nmap-libz-1.3.1 nmap-libpcap-2.10.45 Npcap-1.82 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: iocp poll select

C:\Program Files (x86)\Nmap>nslookup www.afaqwetaan.com
Server: Unknown
Address: 192.168.1.1

Non-authoritative answer:
Name: sites.framer.app
Addresses: 35.71.142.77
          52.223.52.2
Aliases: www.afaqwetaan.com

C:\Program Files (x86)\Nmap>nslookup www.wetaan.com
Server: Unknown
Address: 192.168.1.1

Non-authoritative answer:
Name: www.wetaan.com.cdn.hsgr.net
Addresses: 2a02:4780:43:4c:b4:5564:4647:186a:4213
          2a02:4780:45:b6eb:a400:cec8:84e:836d
          92.113.23.229
          92.113.16.122
Aliases: www.wetaan.com

C:\Program Files (x86)\Nmap>nslookup mashaalkhair.org.sa
Server: Unknown
Address: 192.168.1.1

Non-authoritative answer:
Name: mashaalkhair.org.sa
Addresses: 2a02:4780:11:1082:0:ca8:c6ae:8
          89.117.27.29
```

I opened Command Prompt and navigated to the Nmap installation directory ,Then I ran the command `nmap--version` to check if the tool was properly installed. The terminal showed that Nmap version 7.97 is installed and ready to use.

```
C:\Windows\System32\cmd.exe

C:\Program Files (x86)\Nmap>nmap -sV 35.71.142.77
Starting Nmap 7.97 ( https://nmap.org ) at 2025-06-26 20:54 +0300
Nmap scan report for a0b1d980e1f2226c6.awsglobalaccelerator.com (35.71.142.77)
Host is up (0.037s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp    open  ssl/http
2 services unrecognized despite returning data. If you know the service/version, please submit the following fingerprints at https://nmap.org/cgi-bin/submit.cgi?new-service
!
=====NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)=====
SF:Port80-TCP-V=7,97K1=72D=5/26KTime=685089522P=i686-pc-windows-windows%r(
SF:GetRequest,F6,"HTTP/1.1,0x20308vx20Permanentvx20Redirect\r\nAlt-Svc:vx2
SF:0h3=\":443\";vx20ma=2592000\r\nDate:vx20Thu,vx2026vx20Junvx202025vx2017
SF::54:25vx20GMT\r\nLocation:vx20https:///r\nServer:vx20Framer/220026c\r\n
SF:nStrict-Transport-Security:vx20max-age=31536000\r\nX-Content-Type-Optio
SF:ns:vx20nosniff\r\nContent-Length:vx200\r\n\r\n\"%r(RSPRequest,67,\"HTTP/1.1vx20400vx20Bad
SF:/1.1,0x20308vx20Permanentvx20Redirect\r\nAlt-Svc:vx20h3=\":443\";vx20ma
SF:=2592000\r\nDate:vx20Thu,vx2026vx20Junvx202025vx2017:54:25vx20GMT\r\nLo
SF:cation:vx20https:///r\nServer:vx20Framer/220026c\r\nnStrict-Transport-S
SF:ecurity:vx20max-age=31536000\r\nX-Content-Type-Options:vx20nosniff\r\nC
SF:ontent-Length:vx200\r\n\r\n\"%r(RSPRequest,67,\"HTTP/1.1vx20400vx20Bad
SF:vx20Request\r\nContent-Type:vx20text/plain;vx20charset=utf-8\r\nConnect
SF:ion:vx20close\r\n\r\n400vx20Badvx20Request\"%r(FourOhFourRequest,119,\"H
SF:TP/1.1,0x20308vx20Permanentvx20Redirect\r\nAlt-Svc:vx20h3=\":443\";vx2
SF:0ma=2592000\r\nDate:vx20Thu,vx2026vx20Junvx202025vx2017:54:25vx20GMT\r\n
SF:nLocation:vx20https:///r\nServer:vx20Framer/220026c\r\nnStrict-Transport-Security:vx20max-age=31536000\r\n
SF:X-Content-Type-Options:vx20nosniff\r\nContent-Length:vx200\r\n\r\n\"%r(
SF:GenericLines,67,\"HTTP/1.1vx20400vx20Badvx20Request\r\nContent-Type:vx2
SF:0text/plain;vx20charset=utf-8\r\nConnection:vx20close\r\n\r\n400vx20Bad
SF:vx20Request\"%r(Help,67,\"HTTP/1.1vx20400vx20Badvx20Request\r\nContent-
SF>Type:vx20text/plain;vx20charset=utf-8\r\nConnection:vx20close\r\n\r\n400
SF:0vx20Badvx20Request\"%r(SSLSessionReq,67,\"HTTP/1.1vx20400vx20Badvx20Re
SF:quest\r\nContent-Type:vx20text/plain;vx20charset=utf-8\r\nConnection:vx
SF:20close\r\n\r\n400vx20Badvx20Request\"%r(LPDString,67,\"HTTP/1.1vx20400
SF:vx20Badvx20Request\r\nContent-Type:vx20text/plain;vx20charset=utf-8\r\n
SF:Connection:vx20close\r\n\r\n400vx20Badvx20Request\"%r(CSPOptions,67,\"HT
SF:TP/1.1vx20400vx20Badvx20Request\r\nContent-Type:vx20text/plain;vx20cha
SF:rset=utf-8\r\nConnection:vx20close\r\n\r\n400vx20Badvx20Request\"%r(Soc
SF:ks5,67,\"HTTP/1.1vx20400vx20Badvx20Request\r\nContent-Type:vx20text/pla
SF:in;vx20charset=utf-8\r\nConnection:vx20close\r\n\r\n400vx20Badvx20Reque
```

Port 1

```
C:\Windows\System32\cmd.exe
SF:in;\x20charset=utf-8\r\nConnection:\x20close\r\n\r\n400\x20Bad\x20Reque
SF:st");
=====NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)=====
SF:Port443-TCP:V=7.97X=SSLXI=7D=6/26XTime=685D8959P=i686-pc-windows-win
SF:down;\x20GetRequest,122,"HTTP/1.1\x20400\x20Bad\x20Request\r\nAlt-Svc:\x
SF:20h3=*.443*;\x20max=2592000\r\nConnection:\x20close\r\nContent-Type:\x
SF:20text/plain;\x20charset=utf-8\r\nDate:\x20Thu,\x2026\x20Jun\x202025\x2
SF:017:54:31\x20GMT\r\nServer:\x20Framer/220026c\r\nStrict-Transport-Secur
SF:ity:\x20max-age=31536000\r\nX-Content-Type-Options:\x20nosniff\r\nConte
SF:nt-Length:\x2011\r\n\r\nBad\x20Request")\r\n(HTTPOptions,138,"HTTP/1.1\x2
SF:20405\x20Method\x20not\x20allowed\r\nAlt-Svc:\x20h3=*.443*;\x20max=259
SF:2000\r\nConnection:\x20close\r\nContent-Type:\x20text/plain;\x20charse
SF:=utf-8\r\nDate:\x20Thu,\x2026\x20Jun\x202025\x2017:54:31\x20GMT\r\nServ
SF:er:\x20Framer/220026c\r\nStrict-Transport-Security:\x20max-age=31536000
SF:\r\nX-Content-Type-Options:\x20nosniff\r\nContent-Length:\x2018\r\n\r\n
SF:Method\x20not\x20allowed")\r\n(FourFourRequest,122,"HTTP/1.1\x20400\x2
SF:0Bad\x20Request\r\nAlt-Svc:\x20h3=*.443*;\x20max=2592000\r\nConnection
SF::\x20close\r\nContent-Type:\x20text/plain;\x20charset=utf-8\r\nDate:\x2
SF:0Thu,\x2026\x20Jun\x202025\x2017:54:31\x20GMT\r\nServer:\x20Framer/2200
SF:26c\r\nStrict-Transport-Security:\x20max-age=31536000\r\nX-Content-Type
SF:Options:\x20nosniff\r\nContent-Length:\x2011\r\n\r\nBad\x20Request")\r
SF:(GenericLines,67,"HTTP/1.1\x20400\x20Bad\x20Request\r\nContent-Type:\x
SF:20text/plain;\x20charset=utf-8\r\nConnection:\x20close\r\n\r\n400\x20Ba
SF:d\x20Request")\r\n(RTSPRequest,67,"HTTP/1.1\x20400\x20Bad\x20Request\r\n
SF:Content-Type:\x20text/plain;\x20charset=utf-8\r\nConnection:\x20close\r
SF:\r\n\r\n400\x20Bad\x20Request")\r\n(Help,67,"HTTP/1.1\x20400\x20Bad\x20Req
SF:uest\r\nContent-Type:\x20text/plain;\x20charset=utf-8\r\nConnection:\x2
SF:0close\r\n\r\n400\x20Bad\x20Request")\r\n(SSLSessionReq,67,"HTTP/1.1\x20
SF:400\x20Bad\x20Request\r\nContent-Type:\x20text/plain;\x20charset=utf-8\
SF:SF:r\nConnection:\x20close\r\n\r\n400\x20Bad\x20Request")\r\n(LPDSring,67,"
SF:HTTP/1.1\x20400\x20Bad\x20Request\r\nContent-Type:\x20text/plain;\x20c
SF:harsset=utf-8\r\nConnection:\x20close\r\n\r\n400\x20Bad\x20Request")\r\n(S
SF:IPOptions,67,"HTTP/1.1\x20400\x20Bad\x20Request\r\nContent-Type:\x20te
SF:xt/plain;\x20charset=utf-8\r\nConnection:\x20close\r\n\r\n400\x20Bad\x2
SF:0Request");
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 46.70 seconds
C:\Program Files (x86)\Nmap>
```

Port 2

We took the IP address and used Nmap to scan for open ports. Then, we identified the services running on those ports to understand what kind of connections the website allows. This helps us see if the open ports are normal for the site or if there might be potential security issues.

Port Scan and Service Analysis Using Nmap

Website 1: www.afaagwetaan.com IP: 35.71.142.77

- Open Ports:

80 (HTTP): Used for unencrypted web traffic, running Golang net/http server.

443 (HTTPS): Used for encrypted web traffic, running Golang net/http server.

- Analysis:

These ports are typical for a website.

Presence of port 80 means unencrypted traffic is allowed, which may be a security concern.

Recommended to enforce HTTPS and redirect all HTTP traffic to HTTPS.


```

C:\Program Files (x86)\Nmap>nslookup www.wetaan.com
Server: Unknown
Address: 192.168.1.1

Non-authoritative answer:
DNS request timed out.
    timeout was 2 seconds.
Name:   www.wetaan.com.cdn.hstgr.net
Address: 92.113.23.61
        92.113.16.20
Aliases: www.wetaan.com

C:\Program Files (x86)\Nmap>nmap 92.113.23.61
Starting Nmap 7.97 ( https://nmap.org ) at 2025-06-26 21:17 +0300
Nmap scan report for 92.113.23.61
Host is up (0.096s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 8.87 seconds

```

Nmap Scan Report for www.wetaan.com IP: 92.113.23.61

- **Host Status:** Host is up with 0.096s latency.

- **Open Ports:**

80 tcp (HTTP): Unencrypted web traffic.

443 tcp (HTTPS): Encrypted web traffic.

- **Analysis:**

Both ports are typical and necessary for web servers.

Port 80 allows unencrypted traffic, which may be a security risk if not redirected to HTTPS.

No unusual or unexpected open ports detected.

```

C:\Program Files (x86)\Nmap>nmap 92.113.23.61
Starting Nmap 7.97 ( https://nmap.org ) at 2025-06-26 21:17 +0300
Nmap scan report for 92.113.23.61
Host is up (0.096s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 8.87 seconds

C:\Program Files (x86)\Nmap>nmap 89.117.27.29
Starting Nmap 7.97 ( https://nmap.org ) at 2025-06-26 21:29 +0300
Nmap scan report for 89.117.27.29
Host is up (0.20s latency).
Not shown: 962 filtered tcp ports (no-response), 34 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
80/tcp    open  http
443/tcp   open  https
3306/tcp  open  mysql

Nmap done: 1 IP address (1 host up) scanned in 12.33 seconds

C:\Program Files (x86)\Nmap>

```

Nmap Scan Report for <https://mashaelalkhair.org.sa/> IP: 89.117.27.29

- **Host Status:** Host is up (0.20s latency).

- **Open Ports:**

21 tcp (FTP): File transfer service; may pose security risks if not secured.

80 tcp (HTTP): Unencrypted web traffic.

443 tcp (HTTPS): Encrypted web traffic.

3306 tcp (MySQL): Database service; should be protected from direct internet access.

- **Analysis:**

Ports are typical for a web server with database support.

FTP and MySQL open ports could be security concerns if not properly secured.

Recommend restricting MySQL access and redirecting HTTP traffic to HTTPS.

Conclusion

In summary, this task provided valuable hands-on experience in understanding and analyzing network traffic using Wireshark to examine packet details and identify issues, and Nmap to scan open ports and assess security vulnerabilities. This experience enhanced my understanding of network security fundamentals and how to monitor them effectively.